Life Protection · Public Health

JLU-SPH

2026 Gamified Synthetic Biology Popular Science Project "Gene War" (Jiyin Zhanzheng) Planning Proposal

Safeguard life with the heart of science

Respond to public health with the power of collaboration



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I. Basic Information of the Activity

- **Activity Name:** Gamified Synthetic Biology Science Popularization Project "Gene War"
- **Activity Theme:** Utilizing tower defense strategy games as a medium, leveraging gamification analogies to popularize scientific concepts such as CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats), RNA interference, drug resistance, biofilm, and other synthetic biology concepts, achieving a balance between engagement and scientific rigor.
- **Type of Activity:** ☐ Public Service ☑ Science Education ☑ Cultural Promotion ☐ Community Service ☐ Others:
- **Organizers:** JLU-SPH Team and Tsinghua-A Team

II. Purpose of the Activity

1. **Core Purpose:** To develop a non-commercial, student-oriented gamified science popularization project titled "Gene War", which draws on the framework of classic tower defense strategy games to transform abstract and fragmented biological terminologies into game characters, game mechanics, and level-based strategy choices. The project adopts strategic scenarios as an entry point for knowledge learning, aiming to balance entertainment value with scientific accuracy in the practice of science communication.

2. **Specific Objectives:**

- Cognitive/Knowledge Level: Aiming to assist the general public without a background in biology to understand the core logic behind CRISPR, RNA interference, drug resistance, biofilm, and immune evasion; establishing scientific boundaries by distinguishing the fictionalized game analogies from authentic biological mechanisms.
- Behavioral/Skill Level: To guide participants in reflecting upon the mechanisms and limitations of biotechnologies through strategic deduction, comparison of solutions, and level selection within the game.

- Emotional/Attitudinal Level: To lower the threshold for understanding cutting-edge molecular biology concepts, alter the public's stereotypical perception of professional biological terminologies as abstruse and difficult, and stimulate public interest in exploring synthetic biology .
- **3. Expected Outcomes:**
- Coverage: Multi-platform, online, diverse public audience, primarily targeting non-biology major youth and general internet users.
- Satisfaction Goal: Over 90%

Ⅲ、Target Audience

- Core Audience: Cross-platform, non-biology major general public, including adolescents, non-biology major students, and internet users interested in life sciences.

- Target Characteristics: Individuals who have heard of terms such as CRISPR and drug resistance but possess a fragmented or vague comprehension of these concepts, finding it difficult to visualize or compare related biological processes; familiar with the interactive logic of tower defense strategy games, making them more receptive to scenario-based learning through gaming, thus suitable for science popularization initiated from strategic scenario selection.

IV. Time and Venue of the Activity

- Venue: Online multi-platform release

V. Content and Process of the Activity

[Overview of Activity Process]

Phase	Segment Name	Specific Content	Responsible Personnel	Required Materials
Phase	Project Design Phase	Establishment of a gamified content framework, referencing the tower defense game model to transform scientific concepts into game characters, adversary mechanisms, and level designs; creation of mapping documentation.	Planning Team, Scientific Review Team	Project proposal document, science–game mechanics mapping table, intellectual property checklist
Phase	Prototype Development and Internal Testing	Development of prototypes and internal validation, completion of project prototype; scientific review team checks content accuracy, intellectual property compliance, and verifies entertainment value and public value expression; organization of small-scale internal testing to collect experiential feedback and iterate on mechanism documentation.	All volunteer team, scientific review team, internal test volunteers	Playable prototype, internal test questionnaires, documentation of modifications and iterations
Main Phase	Public Beta Release	Multi-platform public release of the “Gene War” science popularization project, guiding the public to experience strategic game levels	Operations and Communications Team	Finalized science popularization project, science interpretation materials, online feedback collection forms
Main Phase	During Operation	Collection and collation of public feedback, gathering of participants’ experiences, recording of public queries and suggestions regarding concept comprehension, gameplay difficulty, and indications of scientific boundaries	Operations Record Team	Online feedback forms, issue recording sheets
Conclusion Phase	Project Review and Archiving	Summary of testing data and public feedback, evaluation of the gamified science popularization practice in terms of entertainment, scientific accuracy, intellectual property, and public value expression, complete archiving of the entire set of project materials	Overall Responsible Person	Review documentation, complete set of original project files

[Detailed Explanation of Each Segment]

Segment 1: Establishment of Gamified Content Framework

- Core Content: Drawing inspiration from the framework of tower defense strategy games for project content design. Translating CRISPR and RNA interference into in-game abilities of plants or defensive units; incorporating mechanisms such as drug resistance, biofilm, and immune evasion as characteristics of enemy monsters, and establishing multi-stage level selection. Developing a science-to-game mechanism mapping table, where each fictional game mechanism simultaneously addresses two questions: How does this setting assist learners in understanding scientific knowledge; and in what aspects does this setting differ from real biological systems.

Segment 2: Prototype Development and Internal Validation Testing

- Core Content: Completion of the project prototype; scientific review team meticulously checks each piece of scientific content to eliminate errors; intellectual property risk review is conducted to ensure the project is entirely non-commercial. Small-scale internal testing is organized to observe whether users can progress through the game levels while proactively focusing on the underlying biological principles. Feedback is used to adjust level difficulty, textual content, and scientific prompt modules.

Segment 3: Cross-Platform Public Release and Science Engagement Experience

- Core Content: Launching the science popularization project across multiple platforms. During gameplay, users will encounter scenarios where original strategies become ineffective, prompting them to actively consider the underlying reasons and navigate to supporting science interpretation modules, which explain the real scientific knowledge corresponding to CRISPR, RNA interference, biofilm, drug resistance, etc. A boundary disclaimer is prominently displayed: This project utilizes analogical science popularization and does not faithfully replicate authentic biological processes.

Phase 4: Collection of Public Feedback

- Core content: Collect public experience feedback through online forms, record user questions, with a primary focus on whether the general public can distinguish between the fictional settings of the game and real scientific concepts, as well as gather suggestions related to content difficulty and the readability of science popularization materials.

Phase 5: Project Review and Archiving

- Core content: Organize all closed beta and public user feedback, and conduct a project review from four perspectives: entertainment value, scientific accuracy, intellectual property (IP) compliance, and expression of public value.

Summarize the design methodology for gamified science popularization, archive all design documents, mapping tables, and prototype materials.

VI. Personnel Configuration and Division of Labor

- Chief Supervisor (1 person): Responsible for the overall coordination of the joint project, overseeing both the JLU- SPH Team and Tsinghua- A Team, managing project progress, and handling major issues such as inter-team communication, copyright matters, and the guidance of public values.
- Planning and Design Team (2-3 members): Responsible for designing the game's world view, characters, levels, and mechanisms, establishing the correspondence between game and scientific concepts, and writing scripts and textual content.
- Scientific Review Team (2-3 members): Review all content for scientific accuracy, compile mapping tables, and clarify the differences between in-game analogies and actual biological concepts.
- Prototype Team (2-3 members): Responsible for the implementation and creation of prototypes based on the project plan.

- Operations, Communication & Documentation Team (2 members): Responsible for multi-platform promotion and publication, collecting online user feedback, organizing questionnaires and issues logs, and archiving the complete set of project materials.
- Intellectual Property & Compliance Specialist (1 member): Verify project intellectual property (IP), ensure non-commercial attributes, mitigate copyright risks, and oversee the expression of public value.

VII. Material Inventory

- Equipment:
  - Computers
- Document and Design:
  - Complete project design documentation
  - Science-to-game mechanics mapping table
  - Intellectual property (IP) inspection checklist
  - User feedback collection forms for closed beta and public versions
  - Project review summary template
- Produced Materials:
  - Finalized prototype of the "Gene War" (Jiyin Zhanzheng) popular science project
  - Supporting visual and textual materials for scientific interpretation

VIII. Risk Contingency Plan

- Content Risk: The gamified interpretation may cause the public to misunderstand biological concepts. All modules in the project will feature prominent boundary reminders; the Scientific Review Team will oversee the entire process; misunderstanding-related issues will be specially collected through feedback channels, with supplementary explanations provided in subsequent iterations.

- Copyright Risk: Art assets and gameplay materials may involve intellectual property (IP). Only compliant resources will be used, maintaining a non-student, non-commercial scope, with complete copyright tracing records implemented.

IX. Effect Evaluation

- Process Record: Statistical analysis of project experience and visit data; measure the proportion of users who access scientific explanation modules after failing "strategic failure" levels; review user feedback forms to determine whether participants can distinguish between in-game analogies and actual biological concepts.
- Satisfaction Assessment: Collect user experience and feedback through online questionnaires to evaluate entertainment value and comprehensibility of knowledge.
- Project Review and Archiving: Consolidate all feedback, conduct a comprehensive review based on entertainment value, scientific accuracy, intellectual property (IP), and expression of public value, develop a reusable gamified science popularization framework, and fully archive all project documentation.